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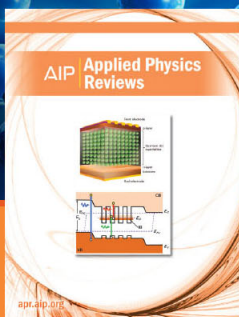
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Spectroscopic ellipsometry of epitaxial ZnO layer on sapphire substrate

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Optical properties of epitaxial ZnO layers have been studied in the spectral region from 1.5 to 5.4 eV using four-zone null spectroscopic ellipsometry. An existing model dielectric function based on excitonic structure near direct band gap has been improved by including a high-energy absorption term. Surface layer, corresponding to the surface roughness, was found to be essential to fit the spectroellipsometric data obtained. Two kinds of samples have been studied: ZnO layers prepared on (0001) and (11 $\bar{2}$ 0)-oriented sapphire substrates. The surfaces of the first ones were found to be more rough. © 2000 American Institute of Physics. [S0021-8979(00)01911-3]

I. INTRODUCTION

Zinc oxide is II–VI semiconductor compound with a large band gap (≈ 3.4 eV) and a strong excitonic structure near the three-dimensional M_0 critical point even in room temperature. It is an attractive material for many applications in optics and optoelectronics. For instance, since a room temperature lasing has been observed in ZnO thin films, the epitaxial ZnO is a promising candidate material for ultraviolet lasers.^{1,2} Because ZnO exhibits acousto-optical properties, it is used as a basic material for integrated acousto-optical devices and surface acoustic wave devices.³ Doped ZnO layers can be used as conducting transparent windows for optoelectronic semiconductor devices.

In view of these applications, there is a need for more accurate data on optical properties of ZnO. Optical properties of the bulk ZnO crystals have been studied using several experimental techniques. A high accuracy minimal deviation method was employed by bond in the transparent spectra region (from 450 nm to 4 μ m).⁴ Far-ultraviolet reflectance spectra were measured by Freeouf for photon energies ranging from 0.6 to 26 eV.⁵

Spectroscopic ellipsometry measurements were performed by Jellison and Boatner⁶ and by Yoshikawa and Adachi⁷ in the spectral interval from 1.5 to 5 eV. Comparing with the other techniques, the spectroscopic ellipsometry has three advantages: (1) two independent quantities (ellipsometric angles ψ and Δ) are measured and, consequently, the real and imaginary part of the complex dielectric function can be obtained directly. (2) The angles ψ and Δ are related to each other by Kramers–Kronig dispersion relations, which are essential for many-parameter determination from measured data. (3) The spectroscopic ellipsometry is very sensitive to ultrathin films and surfaces.

The optical properties of thin ZnO layers are usually determined from transmittance, reflectance, and photoluminescence.^{8,3} A strong excitonic structure was ob-

served by Muth *et al.* in single crystal epitaxial films prepared by pulsed laser deposition.⁹

This article deals with the optical and structural investigation of epitaxial ZnO layer on sapphire substrate. Experimental procedure and sample preparation details are presented in Sec. II. The model including the surface layer and the model dielectric function (MDF) is presented in Sec. III. The experimental results are discussed in Sec. IV.

II. EXPERIMENTAL DETAILS

A. Sample preparation

ZnO films were prepared by rf magnetron sputtering on a single-crystal sapphire substrate. Crystal orientations of sapphire were (1120) [A surface] and (0001) [C surface]. Details of the sample crystal structure and sample preparation have been reported elsewhere by Igasaki.¹⁰ Temperature of the substrate was elevated to 500 °C. Two couples of the sample were prepared, thin (≈ 150 nm) and thick (≈ 400 nm) using the substrates with both orientations. Table I denotes sample numbers used in the paper. Used rf powers were 50 and 100 W for the thin and thick sample, respectively. The films were deposited in an argon gas atmosphere (pressure 10^{-3} Torr) for 30 min. ZnO crystal structure was studied by reflection high-energy electron diffraction (RHEED). In the case of A-sapphire substrate, the initial single-crystalline growth transformed to a polycrystalline one as an effect of the strain originated from high lattice misfit. In the case of C-sapphire substrate, a fiber structure perpendicular to the surface was detected.

B. Spectroscopic ellipsometry

Optical properties of the ZnO films were studied by spectroscopic ellipsometry. An ellipsometer measures ψ and Δ given by

$$\rho = \frac{r_p}{r_s} = \tan \psi e^{i\Delta}, \quad (1)$$

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TABLE I. Sample numbers.

	A-sapphire (1120)	C-sapphire (0001)
Thick	A1 (367.9 nm)	C1 (456.3 nm)
Thin	A2 (121.6 nm)	C2 (158.0 nm)

where r_p and r_s are amplitude reflection coefficients. A computer controlled four-zone null ellipsometer with PSCA (Polarizer–Sample–Compensator–Analyzer) configuration was employed in the spectral region from 1.5 to 5.4 eV (wavelength region from 230 to 840 nm). A Xe–Hg lamp was used as a light source. A 10 cm double-grating monochromator combined with a photomultiplier was employed as a null detector. Rochon quartz prism polarizer and analyzer were rotated by stepping motors to adjust a minimal intensity. A four total reflection-type Fresnel rhomb made from two fused silica rods was used as an achromatic quarter wave retarder. The four-zone averaging for two compensator azimuth angles $\pm 45^\circ$, gives high precise measurements insensitive to the compensator, polarizer, and analyzer azimuth angle errors and imperfections.¹¹ Absence of the systematic errors is essential for a reasonable determination of the unknown parameters. The measurements were performed under the incidence angles of 70° , 75° , and 80° . The measured data for three incidence angles were fitted simultaneously.

Because ZnO is known as an anisotropic crystal, the anisotropy axis of the epitaxial ZnO layer is perpendicular to the surface. Consequently, the complex ellipsometric ratio ρ defined by Eq. (1) is a function of both dielectric constants for ordinary and extraordinary waves. The ellipsometric measurements are unable to separate both components in this configuration, because the wave propagation directions in the layer change only insignificantly and the resultant dielectric function is a mixture of both $\varepsilon_{\parallel}$ and $\varepsilon_{\perp}$.

III. THEORETICAL MODEL

A. Model dielectric function of ZnO

ZnO is a direct-gap II–VI semiconductor compound with a wurtzite crystal structure. The fundamental absorption edge (≈ 3.4 eV) corresponds to a direct transition from the highest valence band to the lowest conduction band. Optical properties of ZnO near the band edge are strongly affected by the excitonic interaction. A model dielectric function (MDF) has to satisfy two conditions: (1) the model is to be based on the interband transition at the energy-band critical points and (2) the real and imaginary parts of the dielectric function are related each other by Kramers–Kronig dispersion relations.

A model dielectric function of ZnO has been published by Yoshikawa and Adachi.⁷ The function includes three interband transitions A, B, C originating from a split of the valence band due to spin-orbit and crystal field interactions. Each transition is described by three terms: E_0 band gap contribution (three-dimensional M_0 critical point) and discrete and continuum excitonic contributions. The MDF includes 15 unknown parameters. Broadening constants in the

excitonic and the band gap terms were considered to be the same. The model exhibits some discrepancies from our measured data in higher frequency region as will be discussed below.

The spectroscopic ellipsometry of a layered media is very sensitive even to a small layer absorption. The absorption affects significantly interference attenuations. Strong sensitivity to an absorption tail is probably a reason why the model dielectric function of Yoshikawa and Adachi did not fit well to our data. The model can be improved considering the broadening constants of the exciton and gap transitions as distinct parameters, however then the model dielectric function would consists of eight parameters for each transition.

The same fit results were obtained using Holden's model dielectric function including only six parameters.¹² The function was employed also by Jellison and Boatner⁶ to fit their spectroellipsometric data. Such dielectric function is described using Lorentzian approximation for line-shape broadening of the excitonic states and the band edge in the form¹² Eqs. (A7)–(A10):

$$\varepsilon_{\text{ex}}(E) = \frac{A}{2E^2} \left\{ \sum_{n=1}^{\infty} [g_{b,n}(E + i\Gamma_n^{\text{ex}}) - g_{b,n}(i\Gamma_n^{\text{ex}})] + g_u(E + i\Gamma_0) - g_u(i\Gamma_0) \right\}, \quad (2a)$$

$$g_{b,n}(\xi) = \frac{8R}{n^3} \frac{E_n^{\text{ex}}}{E_n^{\text{ex}2} - \xi^2}, \quad (2b)$$

$$g_u(\xi) = -\ln \left(\frac{E_0^2 - \xi^2}{R^2} \right) - \pi \left[\cot \left(\frac{\pi \sqrt{R}}{\sqrt{E_0 - \xi}} \right) + \cot \left(\frac{\pi \sqrt{R}}{\sqrt{E_0 + \xi}} \right) \right] - \frac{1}{2} \sum_{n=1}^{\infty} g_{b,n}(\xi), \quad (2c)$$

$$E_n^{\text{ex}} = E_0 - \frac{R}{n^2}, \quad \Gamma_n^{\text{ex}} = \Gamma_0 - \frac{\Gamma_0 - \Gamma_0^{\text{ex}}}{n^2}, \quad (2d)$$

where A is the amplitude, E_0 and Γ_0 are the direct-band-gap energy and the broadening parameter, R is the effective Rydberg energy, E_n^{ex} and Γ_n^{ex} are the energy and the broadening of the n th exciton state, respectively. The infinite sum in Eqs. (2) was calculated approximately as a sum of first five terms and the higher order terms were neglected. The nondispersive term $\varepsilon_{1\infty}$, A , E_0 , Γ_0 , Γ_1^{ex} , and R are fitted parameters.

The fit based on the later model dielectric function (2) properly describes the transparent region (below 3 eV), but some small discrepancies appear in the higher energy region (from 4 to 5.4 eV). Thus, to improve the data fit, we have included higher energy absorptions above the measured interval. The existence of the high energy absorption bands is evident from far-ultraviolet reflectance measurements reported by Freeouf.⁵ The ZnO spectrum exhibits the most important high energy features at 8.6, 12.5, and 15.2 eV. We have described the effects of this absorption to the real part of the dielectric function by adding the Sellmeier term. Any broadening of the absorption was not included to the model,

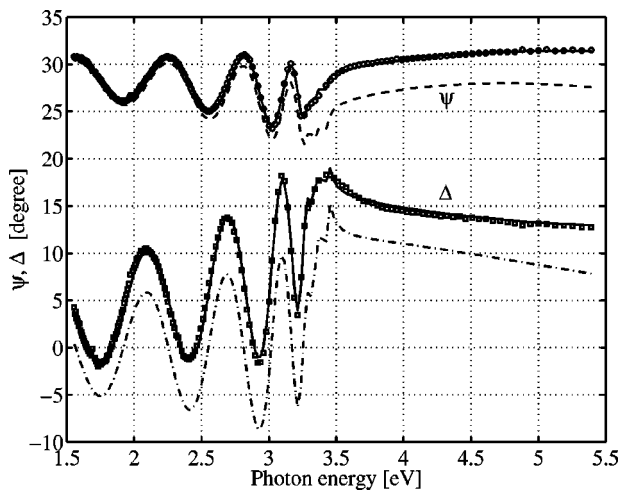


FIG. 1. Spectroellipsometric data of the sample C1 at the angle of incidence of 80°. Measured ψ (circles) and Δ (squares) are compared with the model including surface layer (solid lines) and with the model without surface layer (dashed and dashdot lines).

because it is out of our measured spectra region. The total dielectric function is then taking the following form:

$$\varepsilon(E) = \varepsilon_{\infty} + \varepsilon_{\text{ex}}(E) + \frac{A_1 E_1^2}{E_1^2 - E^2}. \quad (3)$$

The amplitude A_1 and the energy E_1 of the absorption peak are considered as additional fitted parameters.

B. Surface layer

The spectroscopic ellipsometry is very sensitive to the ultrathin films and to the surface quality.¹³ The surface roughness or surface adsorption layers have an important influence on the ellipsometric phase angle Δ . Oscillations of Δ due to the interference effects in non-absorbing layer systems should be symmetric around zero value. Our ellipsometric data shows a strong shift of Δ oscillations in the non-absorbing region (below 3 eV) for all studied samples. The highest shift was observed for the sample C1 (Fig. 1). This shift can be explained by the surface layer existence. The surface layer is probably due to the surface roughness. The surface roughness effects on the ellipsometric data were discussed in details by Ohlídal and Franta.^{14,15} The microrough surfaces are usually modeled by ultrathin surface films of the effective media. In our article, the Bruggeman effective medium approximation (EMA) of the mixture of ZnO and void has been used. The optical properties of the surface layer ε_{ef} are described by the formula:¹⁶

$$0 = (1-p) \frac{\varepsilon - \varepsilon_{\text{ef}}}{\varepsilon + 2\varepsilon_{\text{ef}}} + p \frac{1 - \varepsilon_{\text{ef}}}{1 + 2\varepsilon_{\text{ef}}}, \quad (4)$$

where ε is the dielectric function of ZnO and p is the void volume fraction parameter.

Figure 2 schematically shows the model structure including the surface layer. The Al_2O_3 substrate refractive index was taken as a table value from the Palik handbook.¹⁷

air
$\text{ZnO}_{1-p} + \text{void}_p$ – surface layer
ZnO – MDF
Al_2O_3 – table value [17]

FIG. 2. Model of the structure including the surface layer.

C. Data fit

The obtained spectroellipsometric data were fitted to the appropriate model including several unknown parameters. The considered structure and the model dielectric function were discussed above. The suitable error functions describing the differences between the calculated model and the measured data were minimized to obtain values of the unknown parameters. The Poincare sphere representation of the error function was used in the analysis. Spherical coordinates 2ψ , Δ of the Poincare sphere linearly map the ellipsometric response. Assuming that measured and the calculated points are already close to each other the error function is given by the following formula:

$$\left\{ \sum_{i=1}^N [4(\psi_i^C - \psi_i^M)^2 + (\Delta_i^C \sin 2\psi_i^C - \Delta_i^M \sin 2\psi_i^M)^2] \right\}^{1/2}, \quad (5)$$

where the upper indices C and M represent the calculated and the measured quantity. The described term (5) is divided by a number of fitted data points.

The multidimensional minimization of the error function can be solved by various optimization algorithms different in stability, precision, and computation time.^{18,19} The most robust is the SIMPLEX algorithm, however, its convergence is slow. The faster are less robust gradient-based methods, for example, the Levenberg–Marquardt algorithm. The fits shown below were obtained by combination of both. The real error function exhibits multiple minima. We have obtained a global minimum by searching with randomly chosen

TABLE II. Fit of the spectroellipsometric data of the sample C1 using various models. The fitted thicknesses are shown in brackets [t_{ZnO} (nm); t_{surf} (nm); p (%)]. The error represents the average differences between model and experiment measured on the Poincare sphere.

Model description		Number of parameters	
MDF	surface layer	MDF+structure	Error
MDF4 ^d	no	8+1 (462; —; —)	4.36°
MDF1 ^a	yes	7+3 (459; 16; 34)	1.30°
MDF2 ^b	yes	9+3 (451; 19; 28)	1.18°
MDF3 ^c	yes	6+3 (448; 20; 27)	1.25°
MDF4 ^d	yes	8+3 (456; 17; 36)	0.63°

^aModel of Yoshikawa and Adachi (see Ref. 7), included one transition.

^bModel of Yoshikawa and Adachi (see Ref. 7), included one transition, different Γ for band transition and excitonic structure.

^cHolden formula (see Ref. 12).

^dHolden formula (see Ref. 12) + Sellmeier term.

TABLE III. Fitted parameters for thick and thin ZnO layer deposited on C- and A-oriented sapphire.

	Thick ZnO layer		Thin ZnO layer	
	C-surface C1	A-surface A1	C-surface C2	A-surface A2
t_{ZnO} (nm)	456.3	367.9	158.0	121.6
t_{surf} (nm)	17.1	7.0	7.4	4.3
p (%)	36	31	46	76
$\varepsilon_{1\infty}$	1.22	1.60	1.24	1.85
A (eV ²)	7.84	7.03	7.34	6.97
E_0 (eV)	3.44	3.35	3.38	3.36
Γ_0 (eV)	0.0000	0.0000	0.0036	0.0000
Γ_0^{ex} (eV)	0.047	0.037	0.047	0.039
R (eV)	0.071	0.047	0.053	0.045
A_1	1.35	1.00	1.34	0.74
E_1 (eV)	7.9	8.6	9.4	10.2
error	0.63°	0.47°	0.53°	0.36°

initial estimates of the parameters from reasonable intervals. The minimization procedure with random initial parameters was repeated approximately 100 times.

IV. RESULTS AND DISCUSSIONS

Figure 1 shows the best fit of the sample C1 data with and without the surface layer. From the data in Fig. 1, it is evident that the surface layer is necessary for a proper spectroellipsometric data fit. The surface layer affects only Δ in the nonabsorbing spectra region (< 3 eV), but both ψ and Δ are affected in the absorbing region. The different sensitivities to the surface layer in the absorbing and the nonabsorbing regions show the importance of the simultaneous fit of the whole spectral region. The comparison of our experimental data fits obtained using the various models is summarized in Table II. The ellipsometric spectra of the sample C1 measured at the angles of incidence of 70°, 75°, and 80° were fitted simultaneously. Obtained fitted parameters are summarized in the first column of Table III. The 17.1 nm thick surface layer of the effective medium, containing 36% of the void was obtained for the sample C1. The complex dielectric function of the ZnO layer was also calculated “point by point” from the measured data. The fitted structure parameters (thicknesses and volume fraction) were used for the calculation. Figure 3 shows the complex dielectric function calculated using the latter method directly from the measured data and using the fitted model dielectric function. A good agreement was obtained in the nonabsorbing region. Smooth real and imaginary parts of the dielectric function in this region show that the structure model was chosen properly. The small discrepancies in the absorbing region probably come from (1) nonideal model dielectric function or (2) a small spectral resolution of our ellipsometer. The model dielectric function includes the Lorentz approximation of line broadening. The Lorentz approximation allows to obtain the MDF formula in a close form and ex-

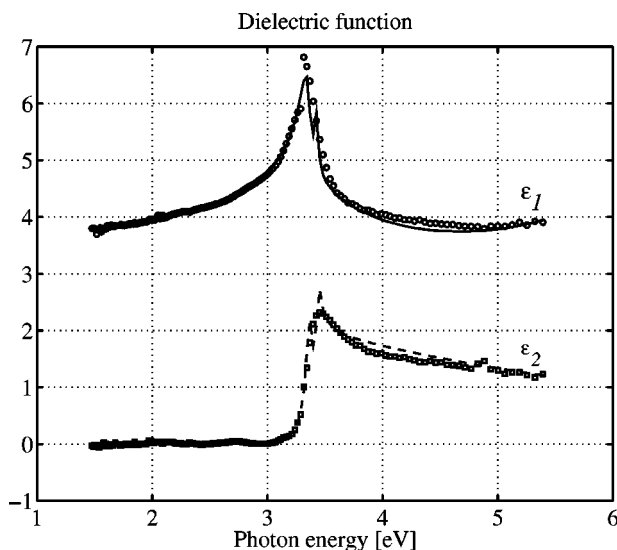


FIG. 3. Real ε_1 and imaginary ε_2 part of the complex dielectric function of the sample C1. Circles and squares represent ε_1 and ε_2 directly calculated from the spectroellipsometric data considering the structure with surface layer. Solid and dashed lines are the real ε_1 and the imaginary ε_2 parts of the fitted model dielectric function.

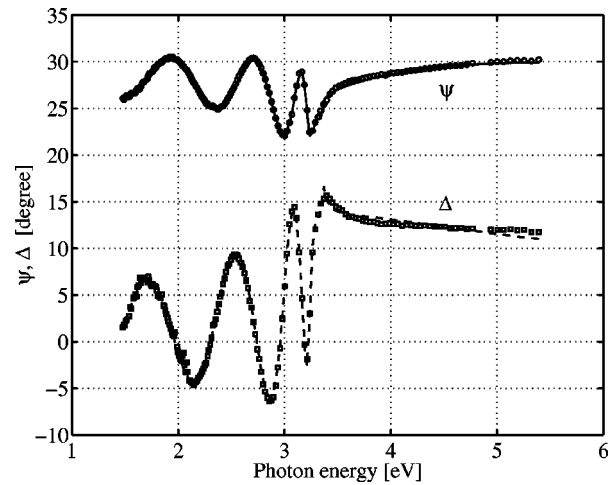


FIG. 4. Spectroellipsometric data of the sample A1 at the angle of incidence of 80°. Measured ψ (circles) and Δ (squares) are compared with the model (solid and dashed lines).

and 80° were fitted simultaneously. Obtained fitted parameters are summarized in the first column of Table III. The 17.1 nm thick surface layer of the effective medium, containing 36% of the void was obtained for the sample C1. The complex dielectric function of the ZnO layer was also calculated “point by point” from the measured data. The fitted structure parameters (thicknesses and volume fraction) were used for the calculation. Figure 3 shows the complex dielectric function calculated using the latter method directly from the measured data and using the fitted model dielectric function. A good agreement was obtained in the nonabsorbing region. Smooth real and imaginary parts of the dielectric function in this region show that the structure model was chosen properly. The small discrepancies in the absorbing region probably come from (1) nonideal model dielectric function or (2) a small spectral resolution of our ellipsometer. The model dielectric function includes the Lorentz approximation of line broadening. The Lorentz approximation allows to obtain the MDF formula in a close form and ex-

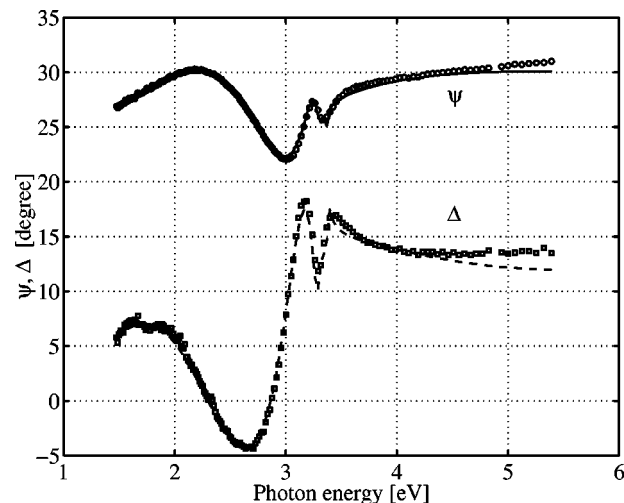


FIG. 5. Spectroellipsometric data of the sample C2 at the angle of incidence of 80°. Measured ψ (circles) and Δ (squares) are compared with the model (solid and dashed lines).

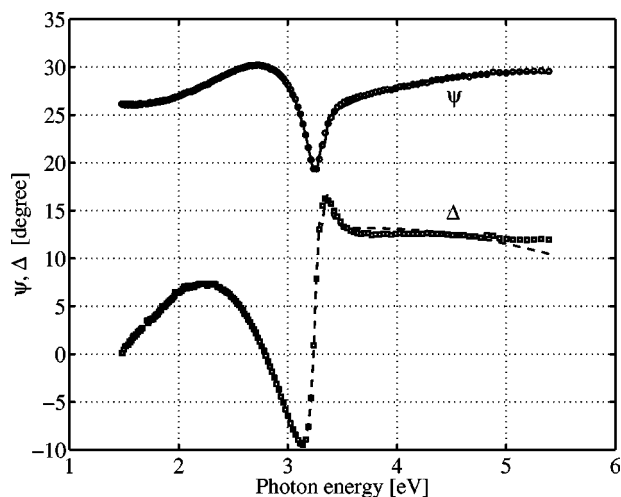


FIG. 6. Spectroellipsometric data of the sample A2 at the angle of incidence of 80°. Measured ψ (circles) and Δ (squares) are compared with the model (solid and dashed lines).

actly satisfying Kramers–Kronig relations. This approximation decreases the accuracy of the obtained parameters Γ_0 , Γ_0^{ex} , and R in our fit.

Figure 4 shows measured data of the sample A1 and their fit. Obtained fitted parameters are summarized in Table III. The surface layer thickness of 7.0 nm and the void volume fraction of 31% were obtained. The surface layer is considerably thinner than for previous sample. Consequently, we can conclude that ZnO layer on A-sapphire is described by higher surface quality than ZnO on C-oriented sapphire. This is due to different crystalline structures discussed in Sec. II. Figures 5 and 6 show the spectroellipsometric data and their fit for thinner ZnO layer (samples C2 and A2). The fitted parameters are summarized in Table III. The obtained surface layer of ZnO on A-sapphire is again thinner than that of ZnO layer on C-sapphire. Figure 7 shows the complex dielectric functions calculated directly from measured data (“point by point”) considering the obtained structure parameters of A1, C2, and A2 samples. The three complex dielectric functions coincide in spite of the surface structure difference.

Several models have been developed for the surface roughness modeling.¹⁴ The applicability of the models depends on the rms value σ and the autocorrelation length T of the rough surface. In this article, we used the simplest model based on the surface layer of the effective medium. The model describes microrough surfaces with $\sigma \ll \lambda$, $T \ll \lambda$. Thickness of 17.1 nm in the case of thick ZnO layer on C-sapphire is probably near the limit of the microrough model applicability.

V. CONCLUSIONS

The epitaxial ZnO layers were prepared on C and A surface of sapphire substrate by rf magnetron sputtering. The ellipsometric spectra of the layers were measured in the energy range from 1.5 to 5.4 eV. The experimental data were

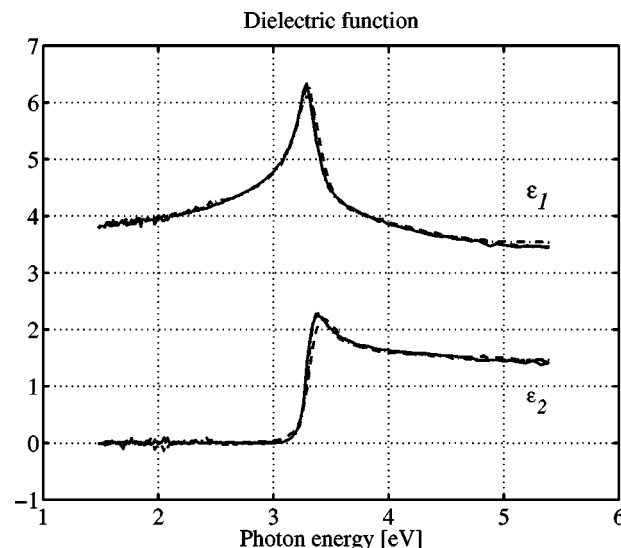


FIG. 7. Real ϵ_1 and imaginary ϵ_2 part of the complex dielectric function of the samples A1, C2, and A2 (solid, dashed, and dashdot lines, respectively) directly calculated from the spectroellipsometric data considering the structure with surface layer.

fitted using the model dielectric function (MDF) of ZnO based on the excitonic state and the three-dimensional M_0 critical point. To improve the data fit, the high energy absorption was included to the MDF. The surface roughness was modeled by a layer of the effective medium. The surfaces of ZnO on C-sapphires were found to be more rough than those of ZnO on A-sapphire. The dielectric function spectra of ZnO were obtained in the investigated spectral interval.

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